

Rohit Rajendra Sharma

+91-9370256399 | rohitsharma250602@gmail.com | India
LinkedIn: linkedin.com/in/rohit-sharma-14aab6293 | GitHub: github.com/rohitsharma2256
Portfolio: rohit-portfolio-nine-azure.vercel.app | LeetCode: leetcode.com/u/rohitsharma250602
Live Product — ComplianceIQ on AWS EC2: Live Demo | Showcase Repo

SUMMARY

Computer Engineering graduate (B.E., 2024) working as a Java backend developer, building and shipping production Spring Boot systems on AWS. Designed and deployed a live multi-tenant SaaS platform that cuts a compliance workflow from 2-3 hours to under 2 minutes across 30+ REST APIs, including an agentic AI assistant and a citation-backed RAG pipeline. Hands-on with JWT-secured REST design, concurrency-safe transactions, PostgreSQL schema modeling, Docker, and production debugging — diagnosed and resolved a live OOM crash via JVM heap tuning. 200+ DSA problems solved in Java.

EXPERIENCE

- Backend Developer — Independent / Freelance** (Client: **DNS Construction Co.**) August 2024 – Present · Remote
Java 21, Spring Boot 3, Spring AI, MCP, PostgreSQL + pgvector, React, Docker, Nginx, AWS
- Delivered **ComplianceIQ** in **Agile** with iterative demos over a **3-4 month engagement** — built for **DNS Construction Co.** as the first client, architected so any CA firm can onboard with fully isolated data; a live **multi-tenant AI SaaS on AWS EC2** spanning **30+ REST APIs** and 9 PostgreSQL entities, cutting a **2-3 hour monthly Excel workflow to under 2 minutes**.
 - Designed a **configurable statutory rules engine** with slabs, ceilings and applicability as versioned data — a government revision becomes a row update, not a redeployment — correctly enforcing the ₹15,000 PF wage ceiling, gross-based ESI eligibility, state-wise Professional Tax and Loss-of-Pay proration.
 - Built an **agentic AI assistant** on **Spring AI** and **Model Context Protocol**: the LLM selects tools, the server owns execution, and multi-step workflows are orchestrated deterministically in code — the model never computes a monetary figure.
 - Engineered a chunked **RAG pipeline on PGVector** (384-dim local MiniLM embeddings, HNSW approximate nearest-neighbour, cosine similarity) for citation-backed statutory answers; intent routing cut per-request tokens **~75%**, and an instant knowledge-update API pushes new rules live to all tenants.
 - Secured the platform with **three-tier RBAC**, service-layer tenant isolation, an **append-only audit trail** with field-level diffs, stateless **JWT + BCrypt**, per-IP token-bucket rate limiting and reCAPTCHA.
 - Automated statutory artefacts — payslips, Form 16, EPFO ECR with pre-export UAN validation, EPF/ESI/TDS challans and audit reports (iText, Apache POI) — plus cron-scheduled deadline alerts over SMTP.
 - Owned **production infrastructure end-to-end**: multi-stage Docker builds (450 MB → 250 MB), **Nginx reverse proxy with TLS termination**, static Elastic IP with DNS A-record mapping, auto-renewing Let's Encrypt certificates; resolved a live **OOM crash** via JVM heap capping and swap provisioning, restoring stable 24/7 uptime.
 - Also built **two backend systems** to strengthen fundamentals — a **timezone-aware course booking engine** with concurrency-safe conflict detection, and an **AI-assisted ticketing backend** with automated priority routing (detailed below).

TECHNICAL SKILLS

Languages: Java (8-21), SQL
Backend & Frameworks: Spring Boot, Spring MVC, Spring Data JPA, Hibernate, REST API Design, JWT Authentication, Spring Security
Databases: PostgreSQL, Oracle, Relational Schema Design
Cloud & DevOps: AWS (EC2, RDS), Docker (multi-stage builds, Compose), Linux, Git, GitHub Actions (CI/CD), Postman
Generative AI (Basics): Spring AI, Model Context Protocol (MCP), RAG, LLM Integration
Core CS: Data Structures & Algorithms, OOP, Multithreading & Concurrency Control, Low-Level Design, Debugging & Root Cause Analysis, DBMS & OS fundamentals

PROJECTS

- Global Class Offering Booking System** Source Code
Java 21, Spring Boot, PostgreSQL, Docker, REST APIs
- Built a scalable booking backend for a global education platform across 15+ REST endpoints, covering the full lifecycle of course offerings, multi-timezone scheduling and seat booking.
 - Eliminated double-booking under concurrent requests using slot-level conflict detection and Spring *@Transactional* integrity, keeping bookings consistent at simultaneous load.
 - Engineered a timezone-aware scheduling engine with the Java Time API (Instant, ZonedDateTime), removing cross-region time-conversion errors for users across geographies.
 - Modeled a normalized PostgreSQL schema (5+ entities) with JPA/Hibernate and optimized foreign-key relationships; shipped via Docker Compose and validated end-to-end in Postman.
- AI Ticketing System — Helpdesk Backend** Source Code
Java 21, Spring Boot, Spring AI, PostgreSQL
- Engineered a self-service helpdesk backend running the complete ticket lifecycle — creation, automatic priority detection (Low/Medium/High), team routing and resolution — modeled on real incident-management workflows.
 - Designed a rule-based classification engine behind a pluggable service layer, allowing drop-in LLM/GenAI classification later with zero refactoring of core services.
 - Cut debugging and root-cause-analysis time using centralized global exception handling and the DTO pattern, producing consistent, observable error responses across all APIs.
 - Modeled the PostgreSQL schema with Spring Data JPA across users, tickets and teams; validated create, assign, prioritize and resolve flows through automated Postman runs.

EDUCATION

B.E. in Computer Engineering — Mumbai University June 2020 – May 2024

CERTIFICATIONS & PROBLEM SOLVING

Certifications: Java Full Stack Developer | DSA with Java | Udemy Java Programming
Problem Solving: 200+ DSA problems solved in Java on LeetCode (arrays, strings, trees, graphs, dynamic programming)